UJJIVAN SMALL FINANCE BANK

Data Scientist / Senior Data Scientist — Hiring Case Study

AI & Data Analytics | Branch Network Expansion Assignment
Data Scientist: 2–5 years experience | Senior Data Scientist: 5–10 years experience

1. Objective

Recommend at least 100 candidate district-level locations for Ujjivan Small Finance Bank's next phase of branch expansion, using only free and publicly available data. The output should be a ranked, tiered, and business-justified set of locations — not a generic model output.

Scope note: districts where Ujjivan already has a branch presence should be excluded. Recommendations must be limited to **new districts** where Ujjivan does not currently operate.

Required branch mix across the 100+ recommendations:

Branch Type	Share	Products
MFI Branches	30%	Group Loan, Gold Loan, Micro LAP
Retail Asset & Liabilities Branches	70%	AHL, MSME, Two Wheeler, CA, SA, TD, TPP, Micro LAP, Gold

Note: MFI and Retail A&L branches serve different customer segments and require different site-selection features (e.g. MFI sites weight rural/semi-urban density, group-lending penetration, and gold-loan demand proxies; Retail sites weight digital/liability readiness, MSME density, and vehicle ownership for two-wheeler demand). Candidates must justify feature choices separately for each branch type, not apply one scoring model uniformly to both.

2. Step 0 — Understand Ujjivan Before You Start

Before touching any data, review Ujjivan SFB's most recent **Investor Presentation** and **Annual Report** (both publicly available at ujjivansfb.in under Investor Relations). Your recommendations must be grounded in Ujjivan's actual footprint and stated strategy — not a generic branch-siting exercise.

Extract and use the following from these documents:

- Current branch network size and state-wise / urban-rural distribution
- Stated growth and expansion priorities (which states/regions Ujjivan is targeting)
- Segment focus mix — microfinance, MSME, affordable housing, FIG
- Any existing site-selection criteria implied by SFB licensing norms (unbanked/underbanked thrust)

3. Free Public Data Sources

Source	What to Extract
RBI DBIE (Database on Indian Economy)	Banking outlet density, deposit/credit data by district
PhonePe Pulse	Digital payment volume/growth by district — proxy for economic activity
Census 2011 / SECC	Population, literacy, household income proxies, urban/rural split by district

RBI list of Banking Outlets / Unbanked Rural Centres	Underserved district identification
PMJDY (Jan Dhan)	Account density by district — financial inclusion proxy
SIDBI / MSME data (optional)	District-level MSME cluster density
NPCI data (optional)	District-level UPI merchant density

Scope: restrict analysis to one or two states (candidate's choice, clearly stated) to keep the assignment feasible within the time budget while still producing 100+ ranked district-level locations.

4. Technical Requirements (50%)

- Source, clean, and join at least 3 of the datasets above at the district level
- Identify and exclude districts where Ujjivan already has an existing branch presence — only new districts should appear in the final ranked list
- Build a composite scoring or ranking model (weighted index, clustering, or ML-based) to rank candidate districts
- Clearly justify feature selection and weighting methodology
- Handle missing or inconsistent district-name mappings across sources (spelling variants, district reorganizations)
- Basic sensitivity check: how stable is the ranking if feature weights shift by 10–20%?
- Submit all code as a well-commented Word or PDF document (see Section 6) — no GitHub link required

5. Functional / Domain Requirements (50%)

- Explain why the chosen features matter specifically for Ujjivan (competition saturation vs. whitespace, digital vs. cash economy readiness, MSME density)
- Segment the 100+ districts into logical clusters (by state, density type, or segment focus) — a flat list is not usable by the business
- Tag each recommended district with its branch type (MFI or Retail A&L) and maintain the required 30% / 70% mix across the final list
- For MFI branches, justify site selection against Group Loan, Gold Loan, and Micro LAP demand indicators (e.g. rural/semi-urban density, gold-loan proxies, group-lending penetration)
- For Retail A&L branches, justify site selection against demand indicators for AHL, MSME, Two Wheeler, CA, SA, TD, TPP, Micro LAP, and Gold products (digital adoption, MSME density, deposit potential, vehicle ownership)
- Flag cannibalization risk — recommended districts too close to existing Ujjivan branches or overlapping with each other
- Recommend a phased rollout: which districts to open first (Tier 1) vs. pipeline (Tier 2/3)
- Note the regulatory angle — RBI SFB norms around unbanked/underbanked area coverage
- Acknowledge data limitations and staleness risk in the recommendation

6. Deliverable Format

Two files, submitted together:

- **A PowerPoint deck** of 10–12 slides, presented as if pitching to Ujjivan's Chief Analytics Officer (CAO)
- **A Word or PDF document containing the full code** (well-commented, with section headers matching the analysis steps) — do not share a GitHub or external repo link

Suggested PPT slide structure:

#	Slide	Purpose
1	Executive Summary	Headline recommendation + tiering (e.g. Tier 1: 20, Tier 2: 40, Tier 3: 40+)
2	Ujjivan Context	Current network snapshot + strategy, from Investor Presentation / Annual Report
3	Methodology	Data sources, scoring approach, feature weights (visual, not code)
4	Ranked Recommendations	Map / table view of top districts, tiered
5	Segmentation & Rationale	Clusters by branch type (MFI 30% / Retail A&L; 70%), state, density; why they fit Ujjivan
6	Cannibalization & Risk Flags	Overlaps with existing branches, data caveats
7	Phased Rollout Plan	Sequencing logic for Tier 1 to Tier 3
8	Appendix	Full 100+ district list with branch type tag; methodology detail

Dark navy / Ujjivan-branded template is a bonus, not mandatory.

7. Evaluation Rubric

Criteria	Weight
Grounding in Ujjivan's actual strategy & footprint (uses report facts, not generic assumptions)	15%
Scoring / ranking methodology rigor & stability	20%
Data handling at scale (100+ districts, clean joins across sources)	15%
Segmentation (MFI/Retail mix), cannibalization logic, phased rollout thinking	25%
Presentation quality — exec summary + appendix, CAO-ready	25%

8. Data Scientist vs. Senior Data Scientist — What We Look For

Data Scientist (2–5 years)	Senior Data Scientist (5–10 years)
Working ranking model with sound feature logic	All DS expectations, plus:
Clean data joins and clear visualization	Proposes how this becomes a repeatable, quarterly-refreshed pipeline
Clear, structured communication in the deck	Flags data reliability / staleness risk explicitly
	Reasons about model governance — how this would be defended under RBI's Model Risk Management guidelines

9. Logistics

- Last date for submission: **10th August 2026**
- Submission: email the PowerPoint deck and the Word/PDF code document to **rajendrareddy.j@ujjivan.bank.in**
- Shortlisting for the interview round will be based entirely on the submitted deck and code